

Tarun Joseph

AI Engineer · Forward Deployed Engineering

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TECHNICAL SKILLS

Languages & Tools: Python, SQL, Git, Docker, FastAPI, Streamlit/Gradio | **LLMs & Agents:** Claude API / Claude Code, OpenAI APIs, LangChain, MCP, RAG, Vector/Graph databases, multi-agent orchestration, tool calling, LLM evaluation & guardrails | **ML & Cloud:** PyTorch, Hugging Face Transformers, SLM fine-tuning, GCP, AWS, Azure, CI/CD (Harness)

EXPERIENCE

- ❖ **Machine Learning Associate | Vector Institute × SafeAlign AI (Fastlane Applied Projects)** *Aug 2026 – Present*
 - Own the governance and policy lane of SafeAlign's security and observability platform for multi-agent AI systems, translating regulatory and policy requirements (pilot: Canada's OSFI E-23, MITRE, EU AI, OWASP) into a knowledge graph that grounds agent-risk decisions.
 - Fine-tuning a small language model (SLM) to reason over that knowledge graph through multi-hop causal traversal, producing traceable, policy-cited compliance decisions.
 - Own deployment and client tuning for the lane across two paths: fully local, on-prem inference for strongly regulated clients, and a cloud variant for less-regulated clients.
- ❖ **AI Program Design & Delivery Intern | Vector Institute** *May 2026 – Aug 2026*
 - Led end-to-end delivery of the Agentic AI Evaluation bootcamp, from onboarding through Showcase Day (9.0/10 participant rating), and resolved technical and operational blockers for 3+ industry teams.
 - Automated the bootcamp lifecycle using the Slack API and Google Apps Script, covering 2 SOPs, team registration, and certificate generation for 13 teams, which saved 5+ hours of prep per bootcamp.
 - Delivered 3 bootcamps, an applied AI cohort, an ICML 2026 speaker series, and a senior-leader AI ROI roundtable with a Director and 5+ PMs, securing 10+ team registrations and curating 4+ speakers.
- ❖ **Machine Learning Engineer | Tata Consultancy Services (TCS)** *January 2023 – May 2025*
 - Deployed GPT-3.5- and Llama 3-powered RAG chatbots and AI agents on TCS's internal EdTech platform, serving 600,000 associates globally.
 - Built a support-ticket automation agent from PoC to MVP using open-source LLMs, with vector-store memory, semantic routing, and API tool calls. Iterated over 3 sprints on stakeholder feedback, saving 5+ hours/week.
 - Led GenAI bootcamps for 10,000+ associates, turning model capabilities into practical workflows for non-technical teams and driving a 13% increase in platform engagement.

PROJECTS

- ❖ **Sentinel: Multi-Agent Prompt-Injection Defense Arena | UofT (ECE1786)** *Sep 2026 – Present*
 - Building a red-team/blue-team system in which an attacker agent injects prompt-injection payloads into tool-using target agents, while a Guardian harness detects, responds, and recovers.
 - Fine-tuning an SLM classifier as the Guardian's detection component, evaluated on attack success rate and task utility against open-source benchmarks.
- ❖ **Support Ticket Automation Agent | Llama 3, Python, Open-Source LLMs** *March 2025*
 - Designed a proof-of-concept (PoC) and built an MVP for an autonomous IT support agent using **Llama 3** with tool-calling capabilities, eliminating repetitive ticket resolution without human intervention.
 - Delivered measurable impact at enterprise scale across a platform used by **600,000 employees**, saving over **five hours** per developer per week.

TEACHING EXPERIENCE

- ❖ **Graduate Teaching Assistant | University of Toronto** *September 2025 – Present*
 - MIE429 AI/ML Capstone Design:** mentored 45 students building AI prototypes and resolved technical blockers in meetings with industry clients.
 - MIE1628 (Cloud-based Big Data Analytics):** Ran tutorials for 100+ students on Data Analytics.

EDUCATION & CERTIFICATIONS

- ❖ **MEng, Industrial Engineering | University of Toronto, Canada** *September 2025 – Present*
 - Relevant Courses:** Deep Learning, Data Science & Statistics, Natural Language Processing, HR Management
 - Emphasis:** Data Analytics & Machine Learning, ELITE | GPA: 3.8/4
- ❖ **B. Tech, Industrial Engineering | College of Engineering Trivandrum, India** *August 2017 – June 2021*
- ❖ **Certifications:** Google: Generative AI Leader Certification (2026) | AWS Certified AI Practitioner (2024) | Microsoft Certified Azure Data Scientist Associate (2024) | AWS Certified Cloud Practitioner (2023)